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ENCLOSURE BOX FOR FIBER OPTIC CABLE

Abstract

An enclosure box assembly for fiber optic cable includes an enclosure box, a backplate, a preterminated fiber optic drop cable preinstalled with the enclosure box, and a fiber optic adapter preinstalled with the enclosure box and optically connected with the fiber optic drop cable. The backplate includes a plurality of latches that are configured to interlock with a complementary latching structure on the enclosure box, and a plurality of slotted countersunk screw holes configured to receive screws for attaching the backplate to a support structure. The backplate includes a first exit opening through a middle portion of the backplate, a second exit opening at a top end of the backplate, and a third exit opening at a bottom end of the backplate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 17/874,028, filed on Jul. 26, 2022, pending, which is a continuation of U.S. application Ser. No. 16/893,278, filed on Jun. 4, 2020, now U.S. Pat. No. 11,397,300, which claims the benefit of U.S. Provisional Application No. 62/857,208, filed Jun. 4, 2019, the disclosures of which are hereby incorporated by reference herein in their entireties.

TECHNICAL FIELD

[0002] This disclosure relates generally to the field of apparatuses for the installation of optical fiber networks. In particular, this disclosure relates to an enclosure box for fiber optic cable and, more particularly, to a customer terminal enclosure box that serves as a termination point for fiber to the home deployments.

BACKGROUND

[0003] As demand for data and connectivity increases, network planners and installers are seeking more efficient and cost-effective deployment options for their fiber to the X (FTTX) rollouts. FTTX comprises the many variants of fiber optic access infrastructure. These include fiber to the home (FTTH), fiber to the premise (FTTP), fiber to the building (FTTB), fiber to the node (FTTN), and fiber to the curb or cabinet (FTTC). The optical FTTP or FTTH network is an optical access network that supplies broadband or ultra-broadband communication services to a number of end users (e.g., services that require data transmission speeds measuring several hundred Mbit/s or even higher).

[0004] An optical FTTP or FTTH network typically includes a fiber distribution hub (FDH), which is typically located in the basement or vault of a multi-dwelling unit building in which the end users reside. A multi-fiber distribution cable typically leads out of the distribution hub to a fiber connection point that is typically located remote from the distribution hub. At the distribution hub, the distribution cable can be connected optically to one or more drop cables. As used herein, a “drop cable” is an optical fiber cable that typically runs between the distribution hub and an apartment or office of an end user. To connect the distribution cable to the drop cables, a number of optical fibers are broken out of the distribution cable and into a same number of single-fiber cables that are each terminated with an optical fiber connector. Each drop cable is then typically terminated at a customer terminal enclosure box, for example, a small customer terminal (SCT), located in the proximity of or inside an apartment or office of an end user. A building may include a single FDH for the entire building, or one FDH per floor, or any combination in between, depending on the fiber connection needs and capacity of the building.

[0005] In some installations, an appropriate length of a preterminated optical fiber drop cable is fed from the FDH to the SCT, and in other installations, the preterminated drop cable is fed from the SCT to the FDH. In either case, the FDH and/or the SCT may be configured to accommodate a slack length of the drop cable when necessary.

[0006] Some conventional SCTs, such as the exemplary SCT **710** shown in FIG. 7, are provided with a preinstalled and preterminated cable **790** and an adapter **792** configured to couple a preterminated end **794** of the cable **790** with another cable (not shown) that is coupled to the

opposite side of the adapter and runs to the end user's Optical Network Terminal (ONT). The SCT **710** includes a base member **712** that is configured to be mounted directly to a wall. In order to secure the base member **712** to the wall, a removable cover (not shown) is removed from the base member **712** and the preinstalled cable is removed to allow access to the mounting features. This lengthens the process for installing the SCT **710** and risks damaging components of the SCT, such as the fiber, the connector, and/or the adapter, when the cable is removed during mounting.

[0007] Therefore, it may be desirable to provide an SCT assembly that permits an SCT to be installed to a wall without accessing an interior of the SCT and without removing preinstalled cable and/or other components. It may also be desirable to provide an SCT assembly that provides multiple exit options from the SCT for the drop cable.

SUMMARY

[0008] According to various embodiments of the disclosure, an enclosure box assembly for fiber optic cable includes an enclosure box, a backplate, a preterminated fiber optic drop cable preinstalled with the enclosure box, and a fiber optic adapter preinstalled with the enclosure box and optically connected with the fiber optic drop cable. The backplate includes a plurality of latches that are configured to interlock with a complementary latching structure on the enclosure box, a plurality of slotted countersunk screw holes configured to receive screws for attaching the backplate to a support structure, and a first exit opening through a middle portion of the backplate, a second exit opening at a top end of the backplate, and a third exit opening at a bottom end of the backplate. The enclosure box is configured to be coupled with the backplate without twisting or turning the enclosure box relative to the backplate to assemble the enclosure box assembly, and the latches and the complementary latching structure are configured to prevent the enclosure box from being removed from the backplate once attached thereto. The second and third exit openings permit a drop cable to be fed through one the second and third exit openings between the enclosure box and the backplate when the enclosure is attached to the backplate and secured via the latches.

[0009] In some aspects, an enclosure box install kit includes a box containing a reel, the aforementioned enclosure box assembly, and a tray containing the enclosure box and the backplate. The fiber optic drop cable is wound on the reel. According to various aspects, the enclosure box install kit includes a cover configured to cover the tray to contain the enclosure box and the backplate in the tray.

[0010] In accordance with various embodiments of the disclosure, an enclosure box assembly for fiber optic cable includes an enclosure box, a backplate, a preterminated fiber optic drop cable preinstalled with the enclosure box, and a fiber optic adapter preinstalled with the enclosure box and optically connected with the fiber optic drop cable. The backplate includes a plurality of latches that are configured to interlock with a complementary latching structure on the enclosure box, and a plurality of slotted countersunk screw holes configured to receive screws for attaching the backplate to a support structure. The backplate includes a first exit opening through a middle portion of the backplate, a second exit opening at a top end of the backplate, and a third exit opening at a bottom end of the backplate.

[0011] In some embodiments, the enclosure box is configured to be coupled with the backplate without twisting or turning the enclosure box relative to the backplate to assemble the enclosure box assembly.

[0012] In various embodiments, the latches are configured as cantilevered arms that extend from a front surface of the backplate and include a finger at the free end of the arms.

[0013] According to some embodiments, the latches and the complementary latching structure are configured to prevent the enclosure box from being removed from the backplate once attached thereto.

[0014] According to various aspects, the enclosure box is a small customer terminal.

[0015] In some embodiments, the second and third exit openings permit a drop cable to be fed through one the second and third exit openings between the enclosure box and the backplate when

the enclosure is attached to the backplate and secured via the latches.

[0016] In various embodiments, the enclosure includes a front wall, a bottom wall, a top wall, two side walls, and a back wall. According to some aspects, the front wall, the bottom wall, the top wall, the two side walls, and the back wall are a monolithic structure of unitary construction such that the front wall is not removable from the walls.

[0017] According to some embodiments, the enclosure box includes first and second wrap guides arranged to direct slack of the fiber optic cable to follow an arcuate path that is greater than or equal to a minimum bend radius of the fiber optic cable.

[0018] According to various embodiments, a front surface of the backplate includes a pair of opposed contoured walls formed by raised ribs extending from the front surface of the backplate on opposite sides of the first opening and extending from the top end of the backplate to the bottom end of the backplate. In some aspects, the walls converge toward one another but are spaced apart from one another at the top end and the bottom end of the backplate to define the second opening and the third opening.

[0019] In accordance with some aspects of the disclosure, an enclosure box assembly for fiber optic cable includes an enclosure box and a backplate. The enclosure box houses a preterminated fiber optic drop cable preinstalled with the enclosure box and a fiber optic adapter preinstalled within the enclosure box and optically connected with the fiber optic drop cable. The enclosure box is configured to be fixedly coupled with the backplate after the backplate is attached to a support structure, and the backplate includes at least one exit opening configured to permit the drop cable to be pass there through.

[0020] According to various embodiments of the disclosure, an enclosure box install kit includes a box containing a reel and the enclosure box assembly according to one of the aforementioned embodiments and a tray containing the enclosure box and the backplate. The fiber optic drop cable is wound on the reel.

[0021] According to some embodiments, a cover is configured to cover the tray to contain the enclosure box and the backplate in the tray.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0022] For a further understanding of the invention, reference will be made to the following detailed description of the invention which is to be read in connection with the accompanying drawing and in which like numbers refer to like parts, wherein:

[0023] FIG. 1 is an exploded perspective view of an exemplary small customer terminal assembly in accordance with various aspects of the disclosure;

[0024] FIG. 2 is a perspective view of a backplate of the exemplary small customer terminal assembly of FIG. 1;

[0025] FIGS. 3A-3C are cutaway views of the exemplary small customer terminal assembly of FIG. 1 in three installation configurations;

[0026] FIG. 4 is a perspective view of an installation kit for the exemplary small customer terminal assembly of FIG. 1;

[0027] FIG. 5 is an exploded perspective view of the installation kit of FIG. 4;

[0028] FIG. 6 is another exploded perspective view of the installation kit of FIG. 4; and

[0029] FIG. 7 is a front view of a conventional small customer terminal assembly.

DETAILED DESCRIPTION

[0030] As a preface to the detailed description, it should be noted that, as used in this specification and the appended claims, the singular forms “a”, “an” and “the” include plural referents, unless the context clearly dictates otherwise.

[0031] FIGS. 1-3B illustrate an exemplary enclosure box assembly or SCT assembly **100** in accordance with various aspects of the disclosure. The SCT assembly **100** includes a customer terminal enclosure box or SCT **110** and a backplate **120**. In some aspects, the backplate **120** includes a plurality of latches **122** that are configured to interlock with a complementary latching structure (not shown) on the SCT **110** such that the SCT **110** is configured to be coupled with the backplate **120** without twisting or turning the SCT **110** relative to the backplate **120** to assemble. In other words, the SCT **110** is pushed toward the backplate **120** to couple the latches **122** with the complementary latching structure. It should be appreciated that the latches **122** may be configured as cantilevered arms **124** that extend from a front surface **128** of the backplate and include a finger **126** at the free end of the arms **124**. It should be appreciated that the complementary latching structure may be any conventional latching structure configured to interlock with the latches **122** without twisting or turning the SCT **110** relative to the backplate **120** to assemble. The latches **122** and complementary latching structure may be configured to prevent the SCT **110** from being removed from the backplate **120** once attached thereto. In some aspects, the latches **122** may be plastic, such as a thermoplastic polymer, for example, acrylonitrile butadiene styrene (ABS).

[0032] As best shown in FIG. 2, the backplate **120** may include a plurality of slotted countersunk screw holes **130** configured to receive screws **132** for attaching the backplate **120** to a wall. The screws **132** may be screwed into studs or anchors configured to receive the screws, as would be understood by persons of ordinary skill in the art.

[0033] The backplate **120** includes an opening **134** through a middle portion of the backplate **120**. The opening **134** is sufficiently sized to feed a drop cable through an opening in the wall behind the backplate **120** that is aligned with the opening **134**. The front surface **128** of the backplate **120** includes a pair of opposed contoured walls **136** formed on opposite sides of the opening **134** and extending from a top end **140** of the backplate **120** to a bottom end **142** of the backplate **120**. As shown, the walls **136** are spaced farthest apart from one another near in the middle of the backplate **120** and converge toward one another, but are spaced apart from one another at the top end **140** and the bottom end **142** of the backplate **120** to define a top exit opening **144** and a bottom exit opening **146**. The walls **136** may be formed by raised ribs extending from the front surface **128** of the backplate **120**. When the SCT **110** is attached to the backplate **120** and secured via the latches **122**, the top and bottom exit openings **144**, **146** permit a drop cable to be fed through either one the openings **144**, **146** between the SCT **110** and the backplate **120**.

[0034] As shown in FIG. 1, the SCT **110** may provide a housing that houses a preinstalled and preterminated drop cable such that an interior chamber of the housing is not accessible during normal use of the SCT **110**. For example, as illustrated in FIG. 1, the front wall **150** of the SCT **110** is integral with the sidewalls of the SCT **110**. Referring now to FIGS. 3A-3C, which shows a cutaway view of the SCT **110** with the front wall **150** removed, the SCT may include the front wall **150**, a bottom wall **152**, a top wall **154**, side walls **156**, **158**, and a back wall **160**. In some aspects, the front wall **150**, the bottom wall **152**, the top wall **154**, the side walls **156**, **158**, and the back wall **160** may be a monolithic structure of unitary construction such that the front wall **150** is not removable from the remaining walls **152-160**. In other aspects, the front wall **150** may be configured as a cover that may be separable from the remaining walls **152-160**. However, in some aspects of a separable front wall **150**, a special keyed tool may be required to remove the front wall **150** in order to avoid unintended access and/or unauthorized access to the interior of the SCT. In such embodiments, the SCT **110** is configured to permit a technician to install the SCT **110** to a wall or surface without having to access the interior of the SCT **110**.

[0035] Referring again to FIGS. 3A-3C, the SCT **110** includes a preinstalled and preterminated drop cable **190**, for example, a preterminated Miniflex® fiber cable, and an adapter **192** configured to couple a preterminated end **194** of the drop cable **190** with another cable (not shown) that can be coupled to the adapter and runs to the end user's Optical Network Terminal (ONT). Although FIGS. 3A-3C illustrate the SCT **110** with two adapters **192**, it should be understood that only a single

adapter is required and thus the second adapter can be omitted. The adapter **192** may be an SC simplex or LC duplex adapter.

[0036] FIGS. **3A-3C** illustrate three installation configurations of the SCT assembly **100**. FIG. **3A** illustrates the drop cable **190** being fed through a center opening **170** of the SCT **110**, between the back wall **160** of the SCT and the front surface **128** of the backplate **140**, and through the bottom exit opening **146** at the bottom wall **142** of the backplate. FIG. **3B** illustrates the drop cable **190** being fed through the center opening **170** of the SCT **110**, between the back wall **160** of the SCT and the front surface **128** of the backplate **140**, and through the top exit opening **144**. FIG. **3C** illustrates the drop cable **190** being fed through the center opening **170** of the SCT **110** and through the opening **134** through the middle portion of the backplate **120**. In each of these installation configurations, the SCT **110** is arranged such that the adapter **192** is oriented vertically so that the cable (not shown) that runs to the end user's ONT will plug into the adapter **192** at the bottom end **142** of the backplate. Of course, the SCT assembly **100** can be rearranged such that the drop cable **190** can be fed in any desired direction, and the adapter **192** can be oriented in any desired direction.

[0037] Referring still to FIGS. **3A-3C**, the SCT **110** may include first and second wrap guides **162**, **164** are arranged to direct the slack fiber optic cable **191** to follow an arcuate path that is greater than or equal to a minimum bend radius of the fiber optic cable **190**. The bend radius of the arcuate path in the storage region of the wrap guides **162**, **164** is generally greater than the minimum permissible bend radius of the fiber optic cable **190** in order to prevent signal losses. For example, as the bend radius of an optical fiber or filament decreases below the minimum permissible bend radius, the incident light energy is not fully reflected internally of the optic filament and, thus, light energy is refracted out of the filament causing a portion of the light energy to be absorbed or lost.

[0038] Referring now to FIGS. **4-6**, the SCT assembly **100** may be provided as part of an install kit **10**. The install kit **10** includes a box **12** containing a reel **14** having the drop fiber optic cable **190** wound thereon between a pair of reel flanges **14a**, **14b**. The box **12** may be configured similar to a pizza box. One of the reel flanges **14a** includes a cutout **16** configured to receive a removable tray **18**. The tray **18** is configured to receive the SCT **110** and the backplate **120**. The tray **18** is configured to receive a removable cover **19** that cooperates with the tray **18** to contain and protect the SCT **110** and the backplate **190** during shipping and storage prior to use. The box **12**, the reel **14**, the tray **18**, and the cover **19** may be disposed after the SCT assembly **100** is installed. For example, in some aspects, the box **12** and the reel **14** may be cardboard, and the tray **18** and the cover **19** may be plastic.

[0039] As discussed above, the SCT **110** includes the preinstalled and preterminated drop cable **190** and the adapter **192**, and a remaining portion of the drop cable **190** that is not contained in the SCT **110** is wound on the reel **14**. The adapter **192** (or plurality of adapters) may assist with maintaining the tray **18** in the cutout **16** during shipping and storage. When a technician is ready to install the SCT assembly **100**, the reel **14** is removed from the box **12**, and the tray **18** is removed from the cutout **16** in the reel **14**. The cover **19**, if present, is then removed from the tray **18**. For example, in some aspects, the tray **18** may include folded-over flaps that form passages to receive the cover **19**, and the cover **19** can be slidably removed from the tray **18** to expose the backplate **120** and the SCT **110**. The backplate **120** can be removed from the tray **18**, as well as any screws and/or anchors that are provided with the install kit **10**.

[0040] Because the drop cable **190** is preinstalled with the SCT **110**, the portion of the drop cable **190** that is not contained in the SCT **110** extends through an opening **17** in the bottom of the tray **18** to the reel **14**, where it is wound thereon. In order for the SCT **110** to be installed, the drop cable **190** has to be freed from the tray **18** and the reel **14**. For example, in some aspects, a slot **15** can be cut in the tray **18**, as shown in FIG. **6**, and the reel **14** can be torn along a perforation **13** to provide openings facilitating freeing of the portion of the drop cable **190** that is not contained in the SCT **110**. In other aspects, the tray **18** and the reel **14** can include preformed openings for freeing the

drop cable **190**. In other aspects, the tray **18** and the reel **14** can include weakened portions that can be removed in order to free the drop cable **190**.

[0041] After the SCT **110** and the backplate **120** are removed from the tray **18** and the drop cable **190** is freed from the tray **18** and the reel **14**, the drop cable **190** is paid out from the reel **14**. That is, the drop cable **190** includes a second preterminated end (not shown) that is fed to a fiber distribution hub (FDH) where the drop cable **190** is optically connected to a fiber optic network. Then, the backplate **120** is mounted to a wall or other structure by screwing the screws **132** through the slotted countersunk screw holes **130** and to the wall or other structure, and the SCT **110** is attached to the backplate **120** by coupling the latches **122** with the complementary latching structure on the SCT **110**.

[0042] It should be understood that various changes and modifications to the embodiments described herein will be apparent to those skilled in the art. Such changes and modifications can be made without departing from the spirit and scope of the present disclosure and without diminishing its intended advantages. It is therefore intended that such changes and modifications be covered by the appended claims.

[0043] Although several embodiments of the disclosure have been disclosed in the foregoing specification, it is understood by those skilled in the art that many modifications and other embodiments of the disclosure will come to mind to which the disclosure pertains, having the benefit of the teaching presented in the foregoing description and associated drawings. It is thus understood that the disclosure is not limited to the specific embodiments disclosed herein above, and that many modifications and other embodiments are intended to be included within the scope of the appended claims. Moreover, although specific terms are employed herein, as well as in the claims which follow, they are used only in a generic and descriptive sense, and not for the purposes of limiting the present disclosure, nor the claims which follow.

Claims

1. An enclosure box assembly for fiber optic cable comprising: an enclosure box; a backplate; a preterminated fiber optic drop cable preinstalled with the enclosure box; and a fiber optic adapter preinstalled with the enclosure box and optically connected with the fiber optic drop cable, wherein the backplate includes a plurality of latches that are configured to interlock with a complementary latching structure on the enclosure box, wherein the backplate includes a plurality of slotted countersunk screw holes configured to receive screws for attaching the backplate to a support structure, wherein the backplate includes a first exit opening through a middle portion of the backplate, a second exit opening at a top end of the backplate, and a third exit opening at a bottom end of the backplate, wherein the enclosure box is configured to be coupled with the backplate without twisting or turning the enclosure box relative to the backplate to assemble the enclosure box assembly, wherein the latches and the complementary latching structure are configured to prevent the enclosure box from being removed from the backplate once attached thereto, and wherein the second and third exit openings permit a drop cable to be fed through one of the second and third exit openings between the enclosure box and the backplate when the enclosure is attached to the backplate and secured via the latches.
2. An enclosure box install kit comprising: a box containing a reel and the enclosure box assembly of claim 1; and a tray containing the enclosure box and the backplate, and wherein the fiber optic drop cable is wound on the reel.
3. The enclosure box install kit of claim 2, further comprising a cover configured to cover the tray to contain the enclosure box and the backplate in the tray.
4. An enclosure box assembly for fiber optic cable comprising: an enclosure box; a backplate; a preterminated fiber optic drop cable preinstalled with the enclosure box; and a fiber optic adapter preinstalled with the enclosure box and optically connected with the fiber optic drop cable, wherein

the backplate is configured to be attached to a support structure, wherein the backplate includes a plurality of latches that are configured to interlock with a complementary latching structure on the enclosure box to secure the enclosure box to the backplate after the backplate is attached to the support structure, and wherein the backplate includes at least one exit opening configured to permit the drop cable to pass there through.

5. The enclosure box assembly of claim 4, wherein the enclosure box is configured to be coupled with the backplate without twisting or turning the enclosure box relative to the backplate to assemble the enclosure box assembly.

6. The enclosure box assembly of claim 4, wherein the latches are configured as cantilevered arms that extend from a front surface of the backplate and include a finger at the free end of the arms.

7. The enclosure box assembly of claim 4, wherein the latches and the complementary latching structure are configured to prevent the enclosure box from being removed from the backplate once attached thereto.

8. The enclosure box assembly of claim 4, wherein the enclosure box is a small customer terminal.

9. The enclosure box assembly of claim 4, wherein the at least one opening permits a drop cable to pass between the enclosure box and the backplate when the enclosure is attached to the backplate and secured via the latches.

10. The enclosure box assembly of claim 4, wherein the enclosure includes a front wall, a bottom wall, a top wall, two side walls, and a back wall.

11. The enclosure box assembly of claim 10, wherein the front wall, the bottom wall, the top wall, the two side walls, and the back wall are a monolithic structure of unitary construction such that the front wall is not removable from the walls.

12. The enclosure box assembly of claim 4, wherein the enclosure box includes first and second wrap guides arranged to direct slack of the fiber optic cable to follow an arcuate path that is greater than or equal to a minimum bend radius of the fiber optic cable.

13. The enclosure box assembly of claim 4, wherein a front surface of the backplate includes a pair of opposed contoured walls formed by raised ribs extending from the front surface of the backplate on opposite sides of a first opening of the at least one opening and extending from the top end of the backplate to the bottom end of the backplate.

14. The enclosure box assembly of claim 13, wherein the walls converge toward one another but are spaced apart from one another at the top end and the bottom end of the backplate to define two additional openings of the at least one opening.

15. An enclosure box install kit comprising: a box containing a reel and the enclosure box assembly of claim 4; and a tray containing the enclosure box and the backplate, wherein the fiber optic drop cable is wound on the reel.

16. The enclosure box install kit of claim 15, further comprising a cover configured to cover the tray to contain the enclosure box and the backplate in the tray.

17. An enclosure box assembly for fiber optic cable comprising: an enclosure box housing a preterminated fiber optic drop cable preinstalled with the enclosure box and a fiber optic adapter preinstalled within the enclosure box and optically connected with the fiber optic drop cable; and a backplate, wherein the enclosure box is configured to be fixedly coupled with the backplate after the backplate is attached to a support structure, and wherein the backplate includes at least one exit opening configured to permit the drop cable to be pass there through.

18. An enclosure box install kit comprising: a box containing a reel and the enclosure box assembly of claim 17; and a tray containing the enclosure box and the backplate, wherein the fiber optic drop cable is wound on the reel.
